YAKEEN

NEET 2.0 2027

Physics | Lecture - 12

Motion in a Straight Line

By- Manish Raj (MR Sir)

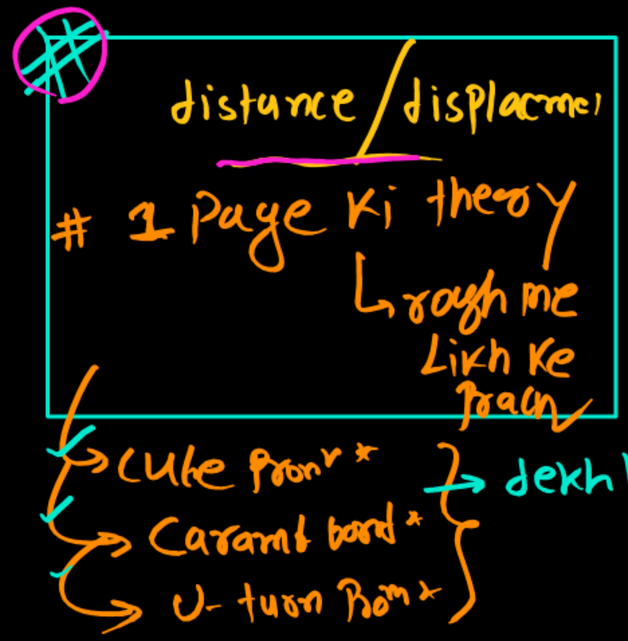


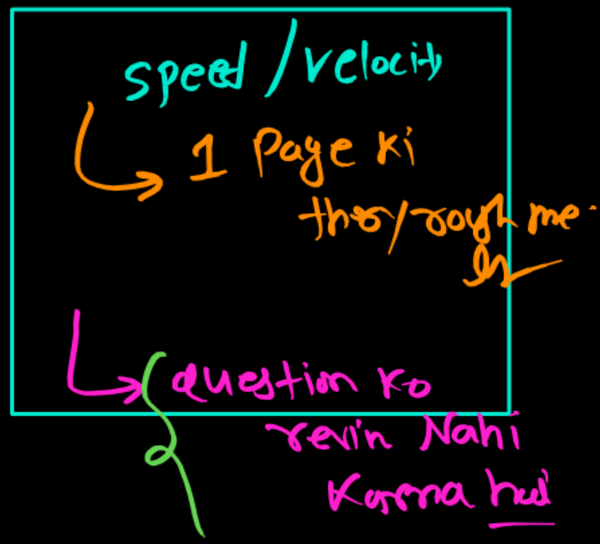
Todays goal

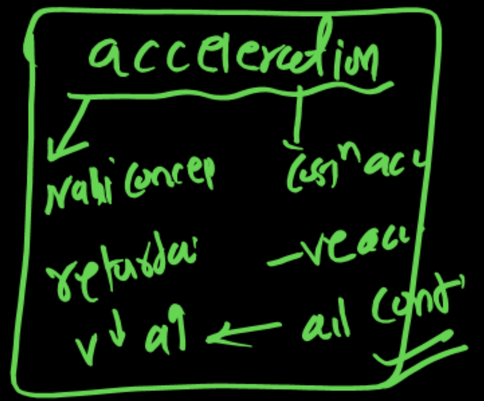
Feel of motion under gravity

H/w goal

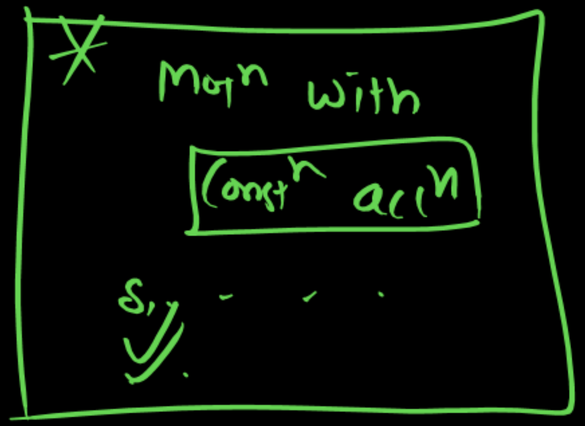
do all H/w given P.P.T


distance / displacement
1 Page ki theory
↳ rough me Likh ke Rakho
↳ Cube prom *
↳ Caramb board *
↳ U-turn Bom *
→ dekh lena


Speed / Velocity
↳ 1 Page ki theory/rough me.
↳ question ko revise Nahi Karna hai!


acceleration
Nahi concept
Constant acc
retarded
-ve acc
v ↓ a ↑
all Const


Integration
diff
Kuch good re.


Math with
Const acc
So,


PhD on
graph.


Math under
gravity
Juggle prop

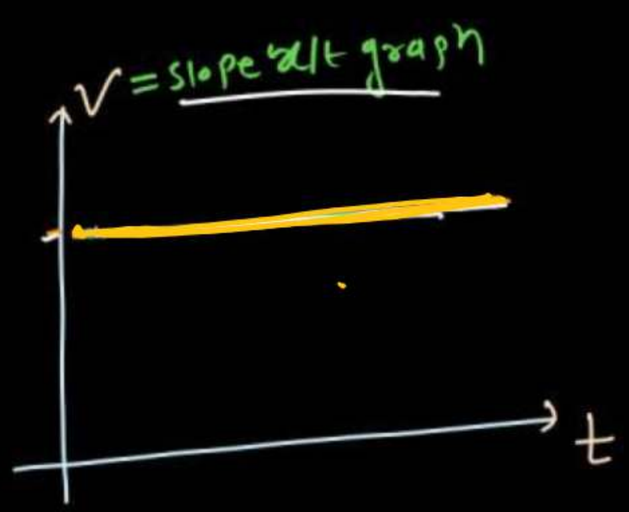
Question H/w kal discuss H/w kal karke aana

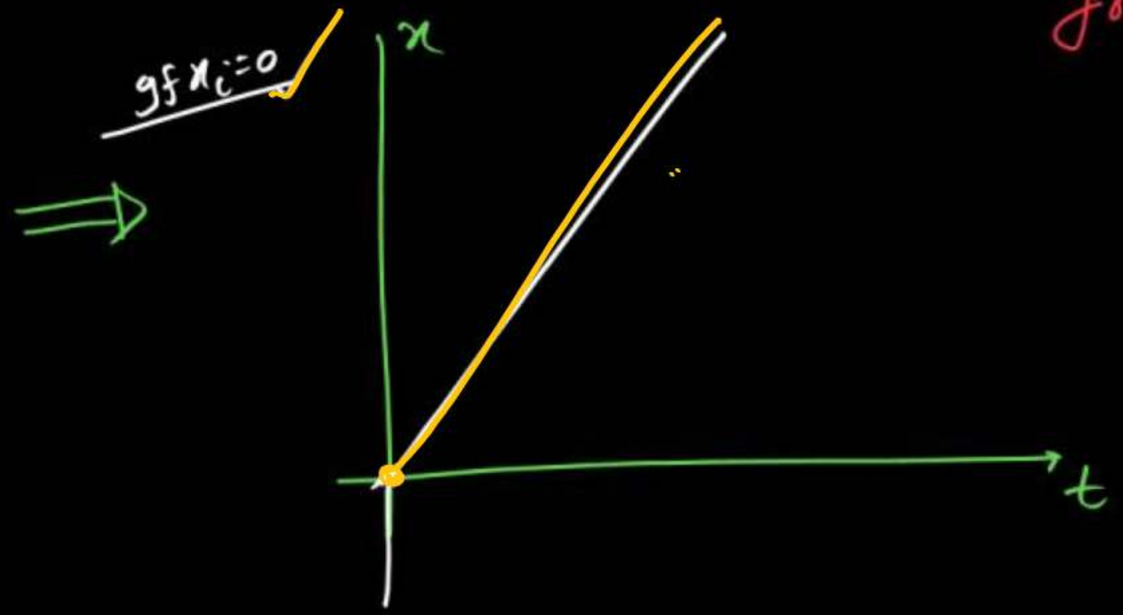
A car starts from rest and moves with uniform acceleration a on a straight road from time $t = 0$ to $t = T$. After that, a constant deceleration brings it to rest. In this process the average speed of the car is

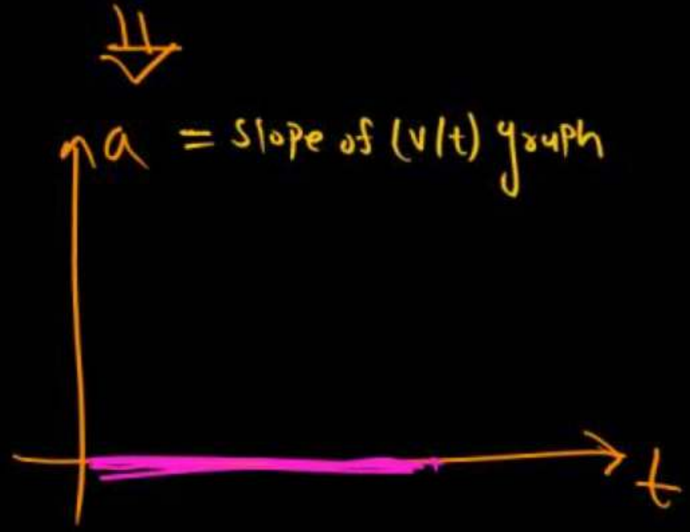


1. aT
2. $\frac{aT}{4}$
3. $\frac{3aT}{2}$
4. $\frac{aT}{2}$

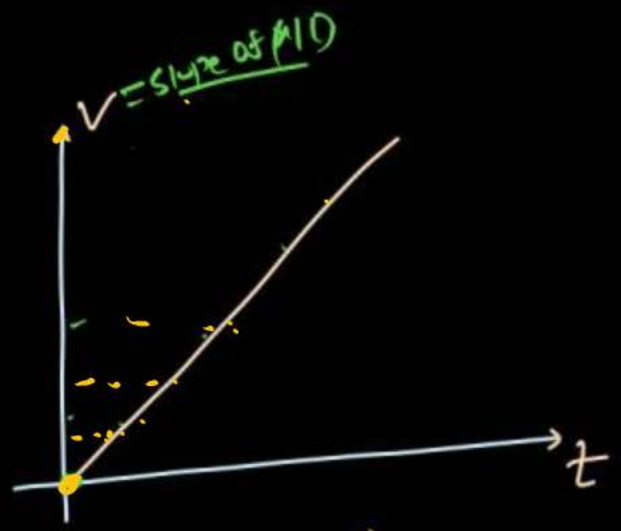
(Q) (SD) Convert velocity time graph into Position-time and acc^{n}-time graph.



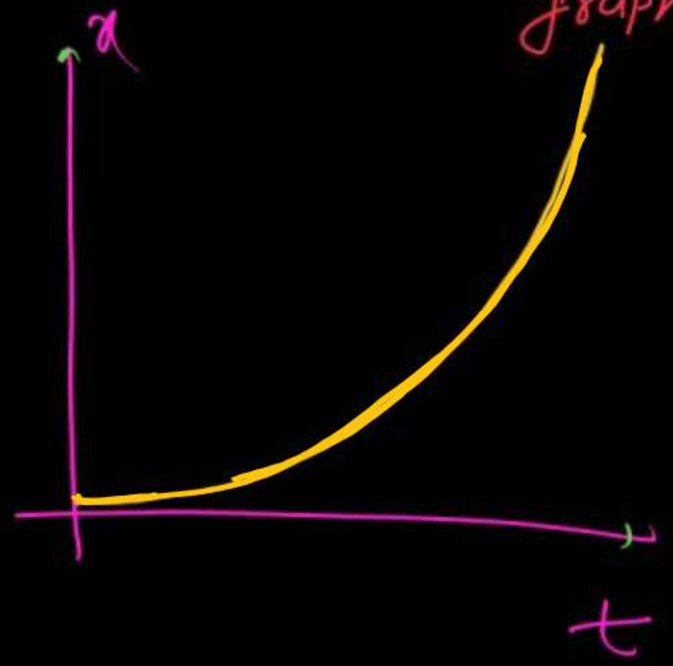


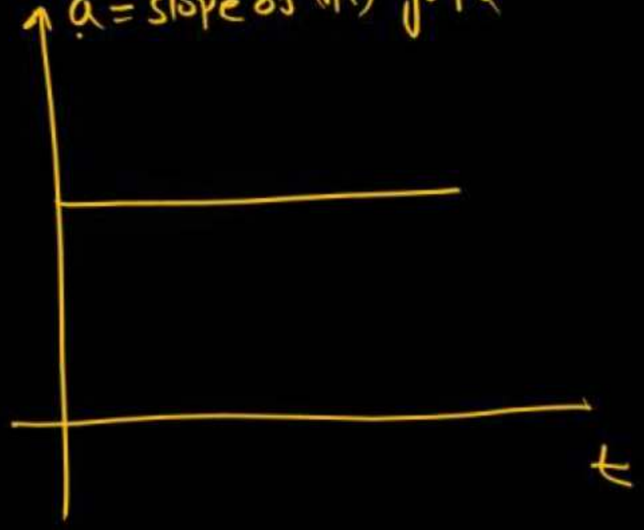


(Q) (57) Convert velocity time graph into Position-time and acc^{n}-time graph.

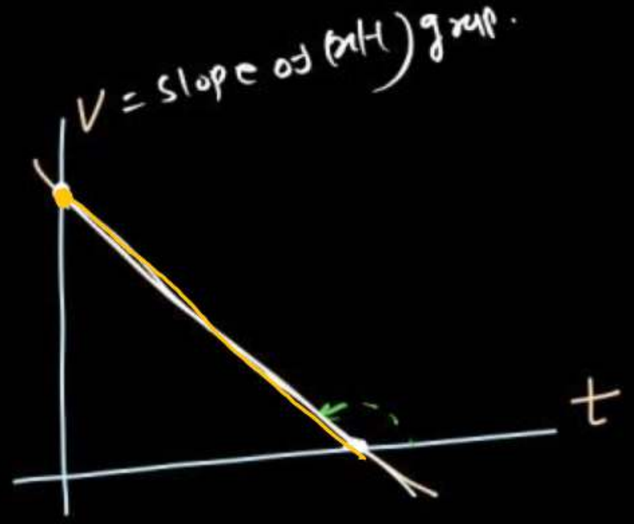




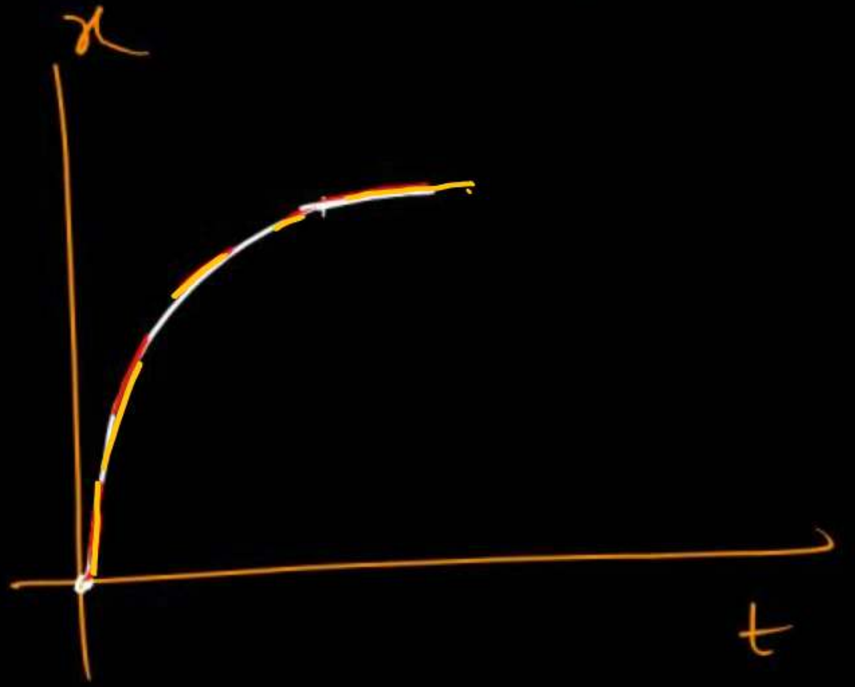


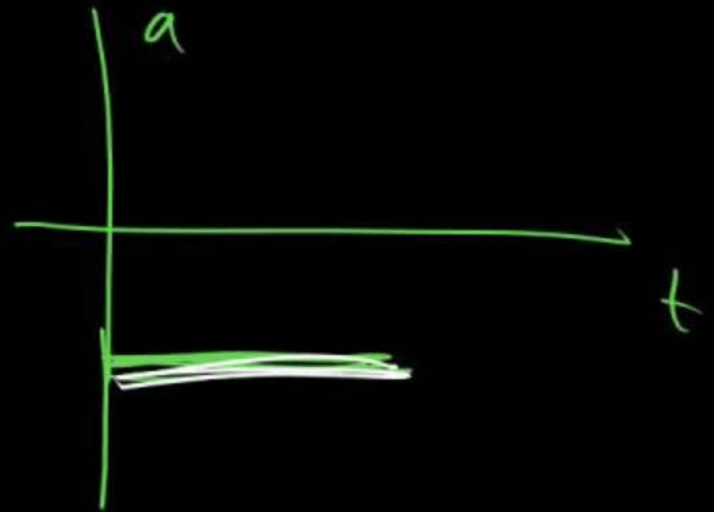


58 (Q) Convert velocity time graph into Position-time and acc^{n}-time graph.



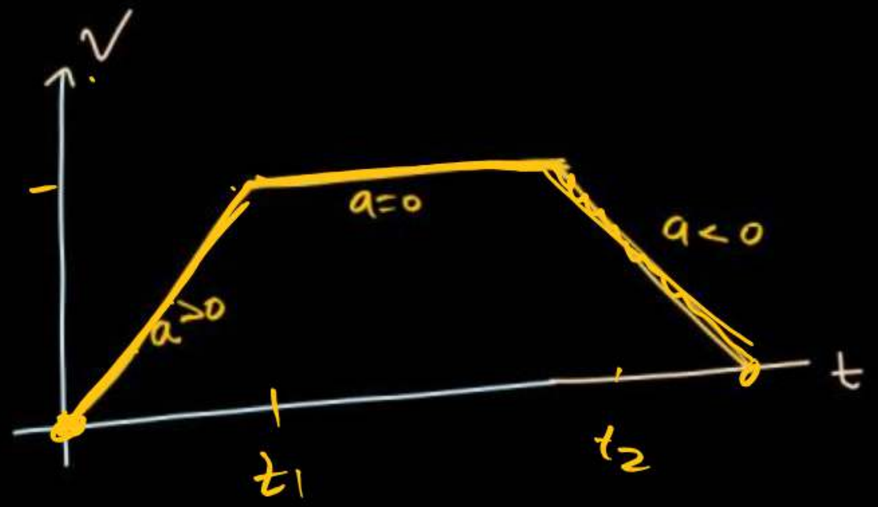


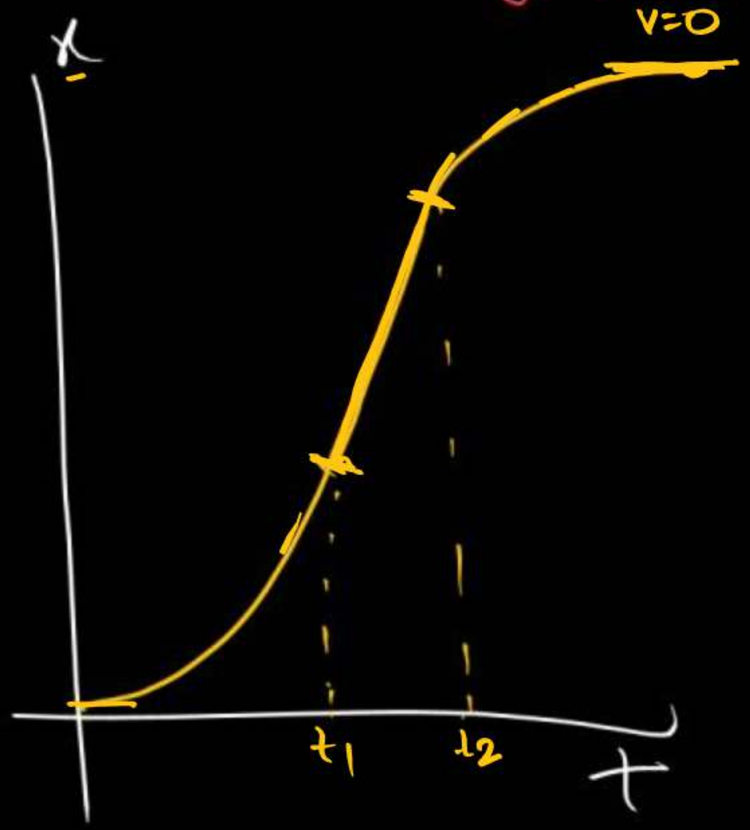


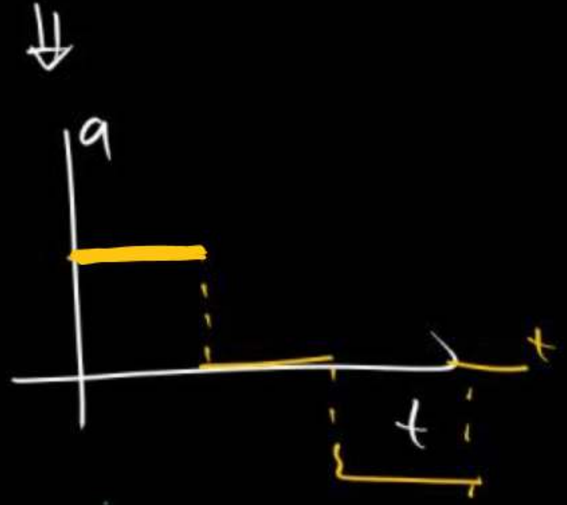


(Q) Convert velocity time graph into Position-time and acc^{n}-time graph.

(59)

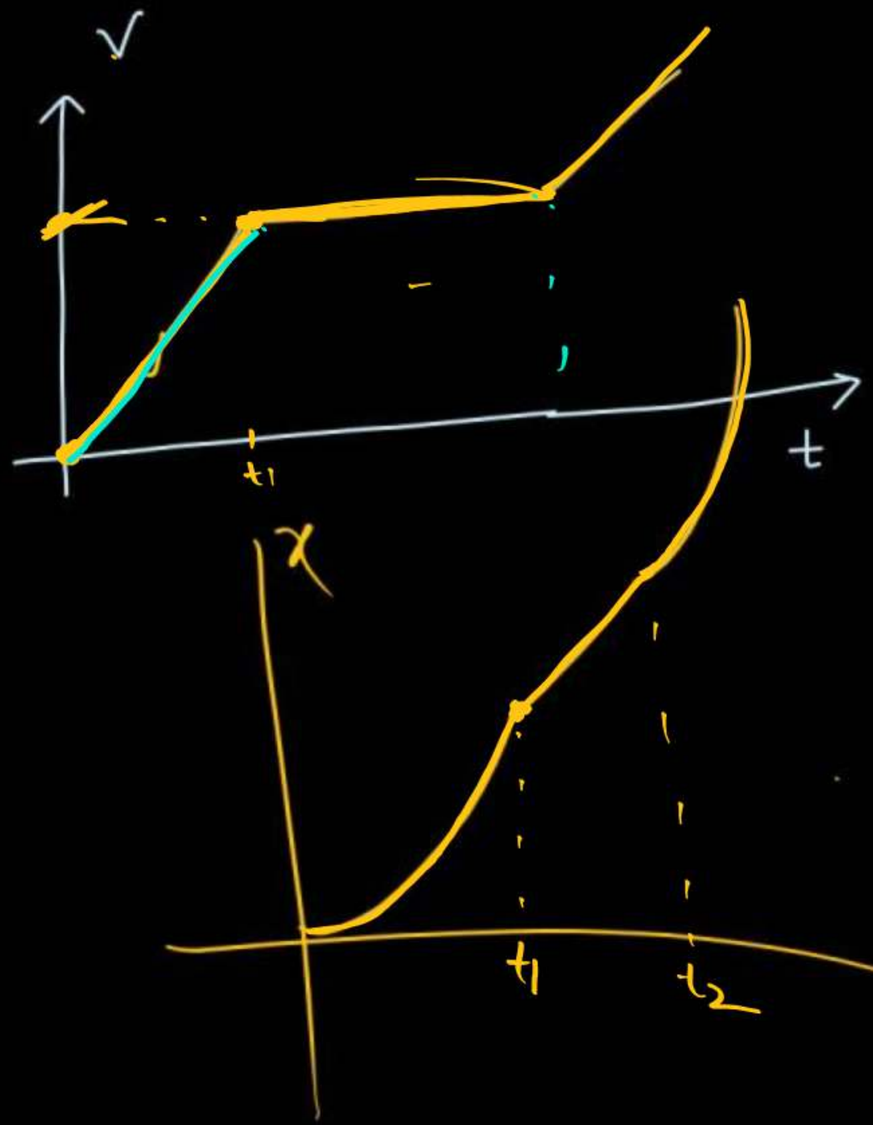


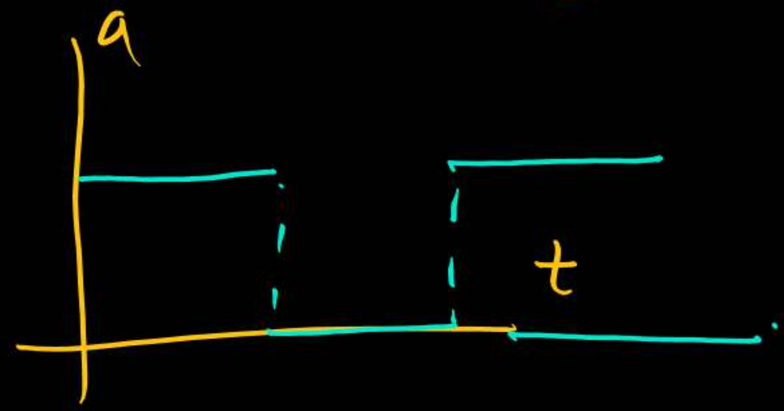




(Q) Convert velocity time graph into Position-time and acc^{n}-time graph.

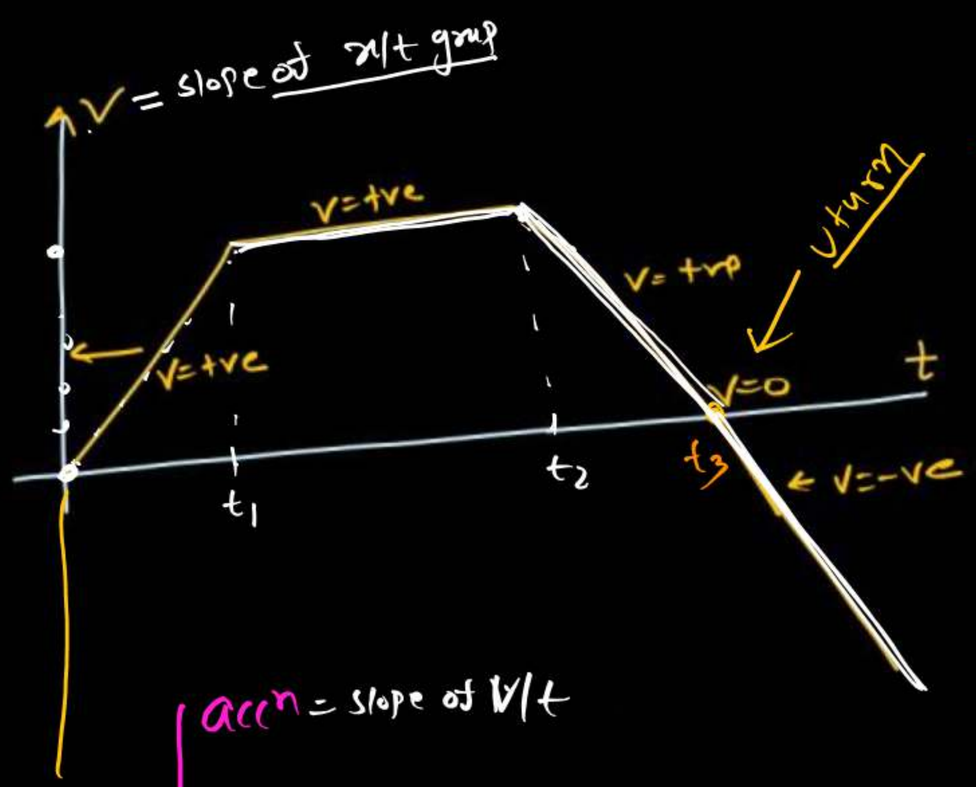
60



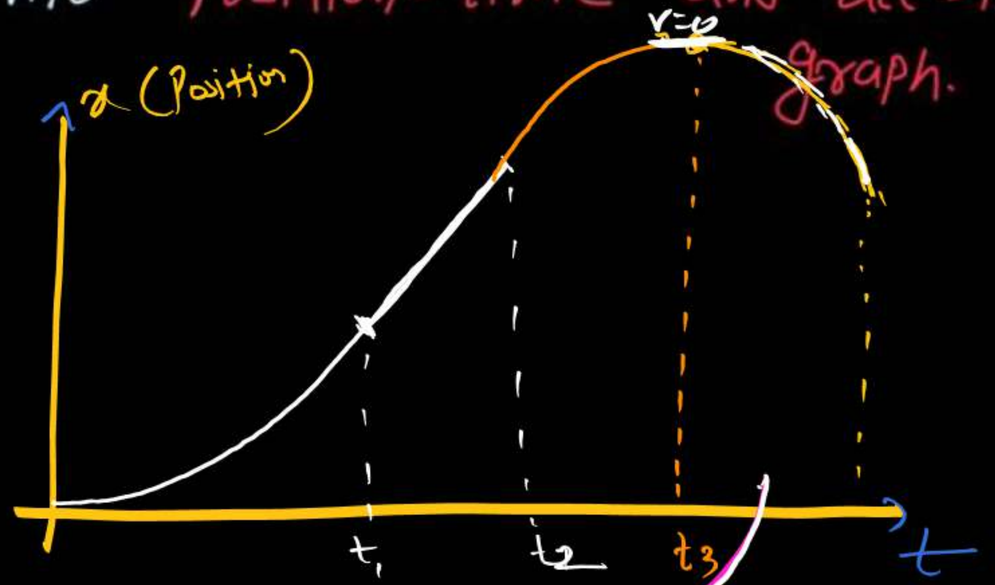


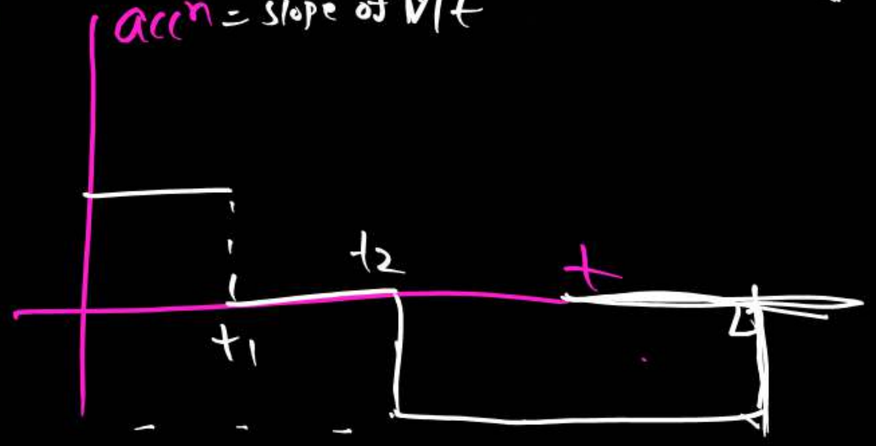
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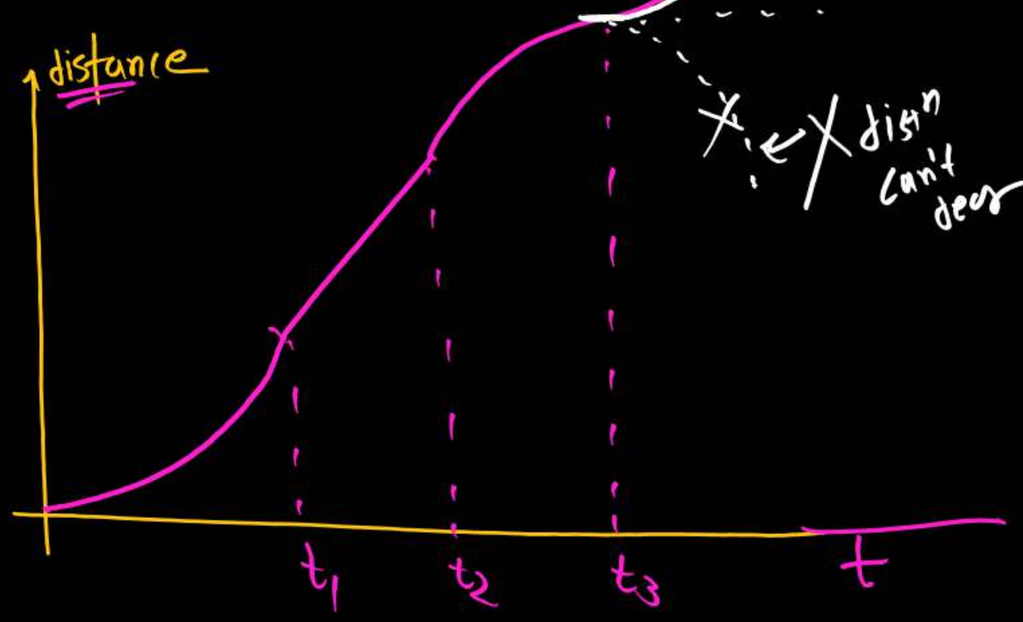
61



x/t

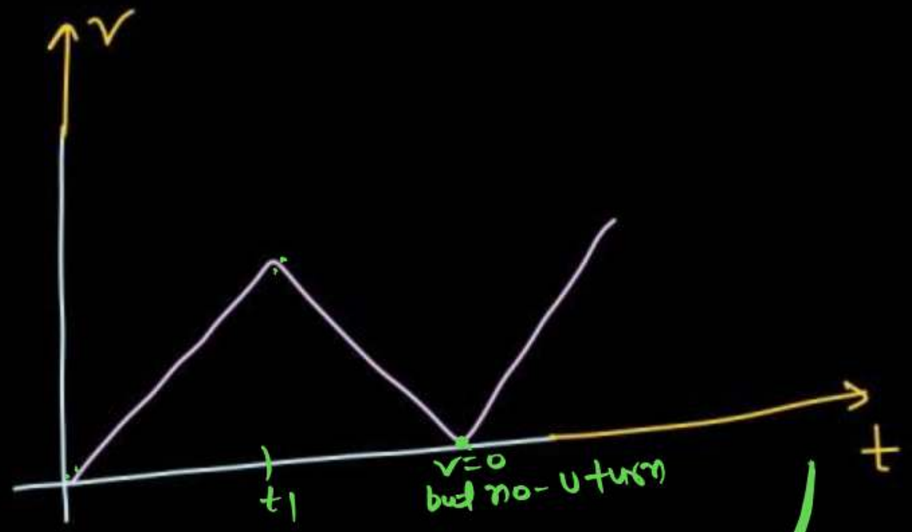


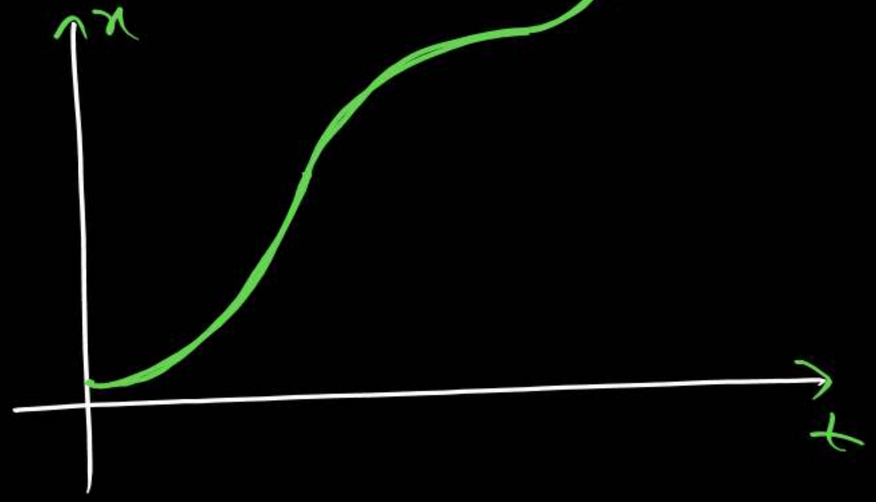


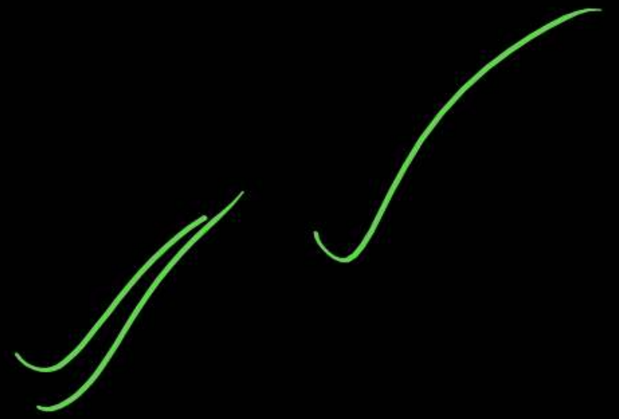


(Q) Convert velocity time graph into Position-time and acc^{n}-time graph.

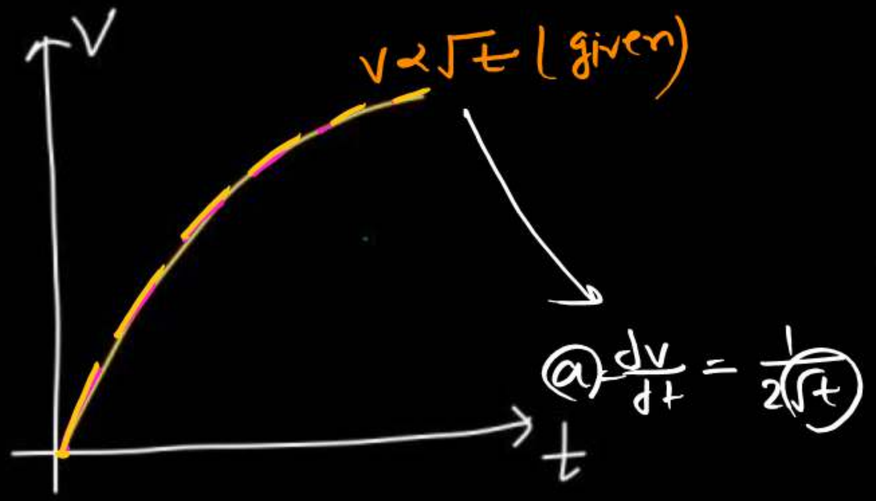
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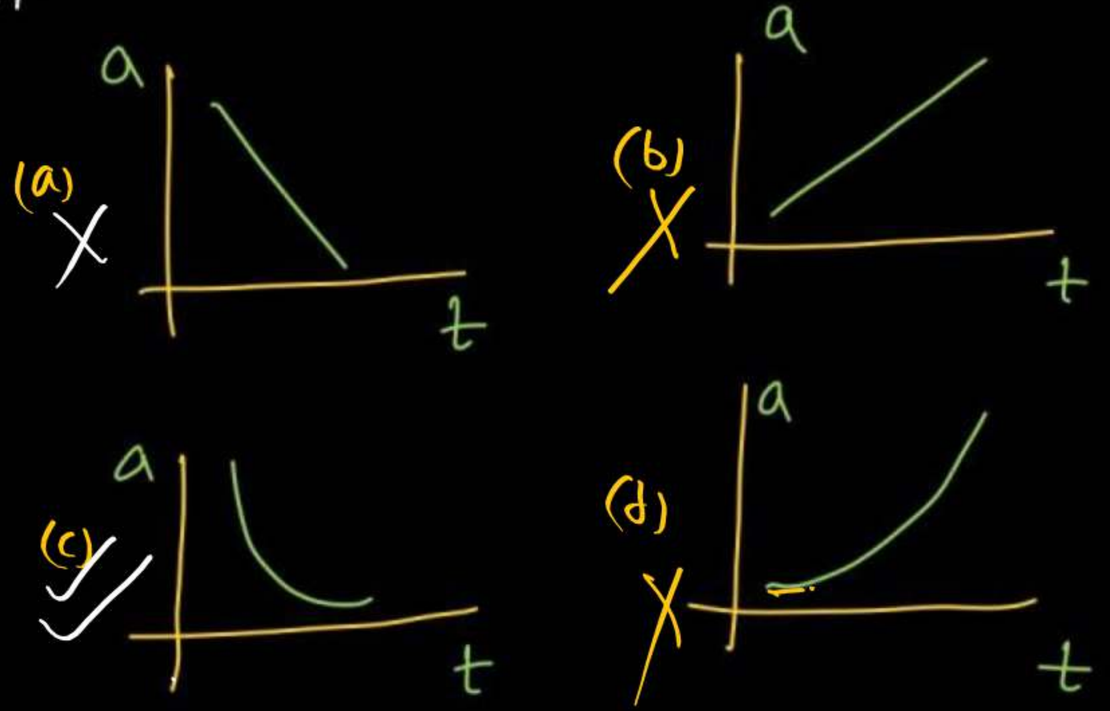






63 for given velocity-time graph which of the following acc-time graph is correct



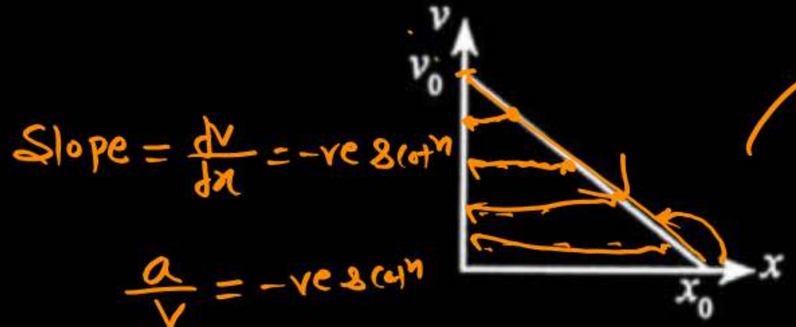


Slope of v/t = accn = +ve & ↓

Question 65

The given graph shows the variation of velocity with displacement. Which one of the graph given below correctly represents the variation of acceleration with displacement?

[JEE Adv. 2015]



Slope $= \frac{dv}{dx} =$ -ve & constant

$\frac{a}{v} =$ -ve & constant

$a = v(-\text{ve \& constant})$ (fixed)

Slope $= -K$ (let) → Method

$y = mx + c$

$v = -Kx + v_0$ — (1)

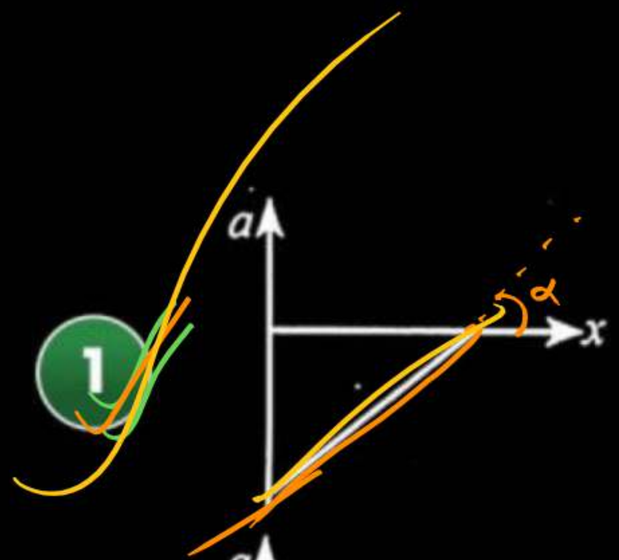
$a = v\frac{dv}{dx}$

$a = (-Kx + v_0)\frac{d(-Kx+v_0)}{dx}$

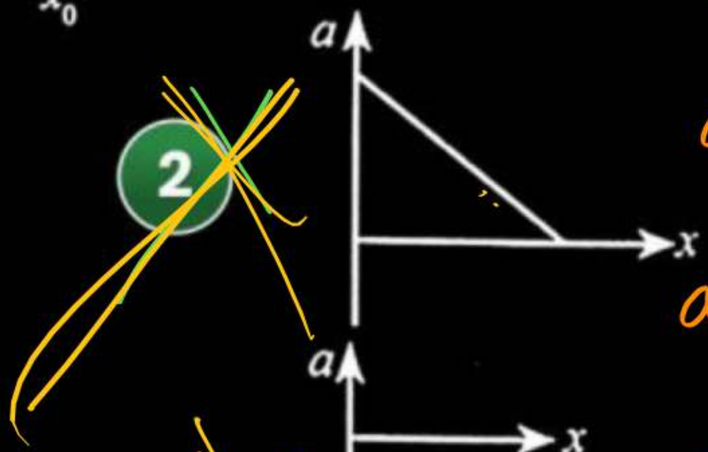
$a = -Kx + v_0\,[-K \times 1 + 0]$

$a = +K^2x - v_0K$

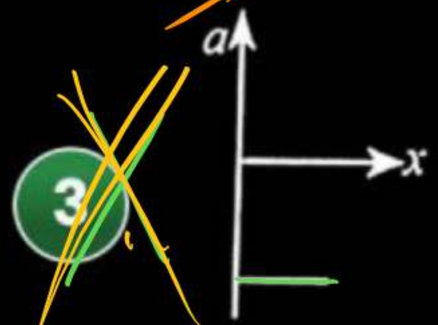
1



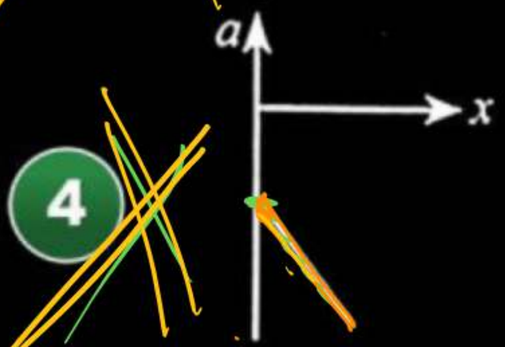
2



3

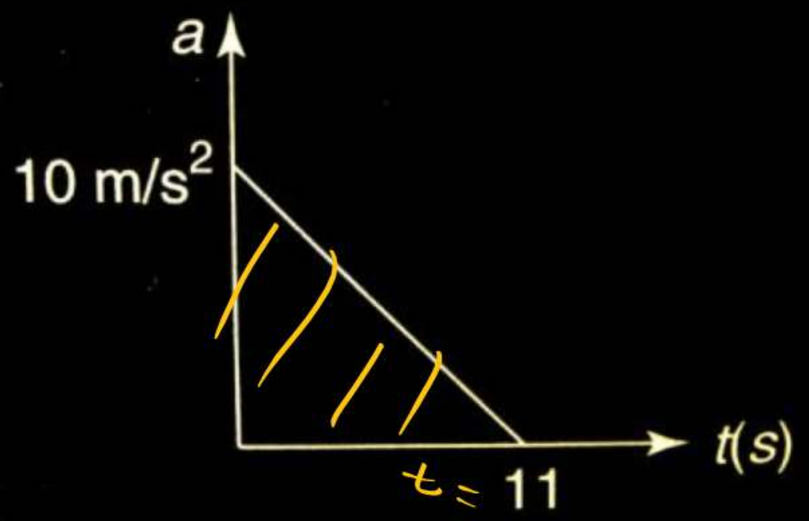


4



Question 66

A particle starts from rest. Its acceleration (a) versus time (t) is as shown in the figure. The maximum speed of the particle will be **[IIT-JEE (Screening) 2004]**

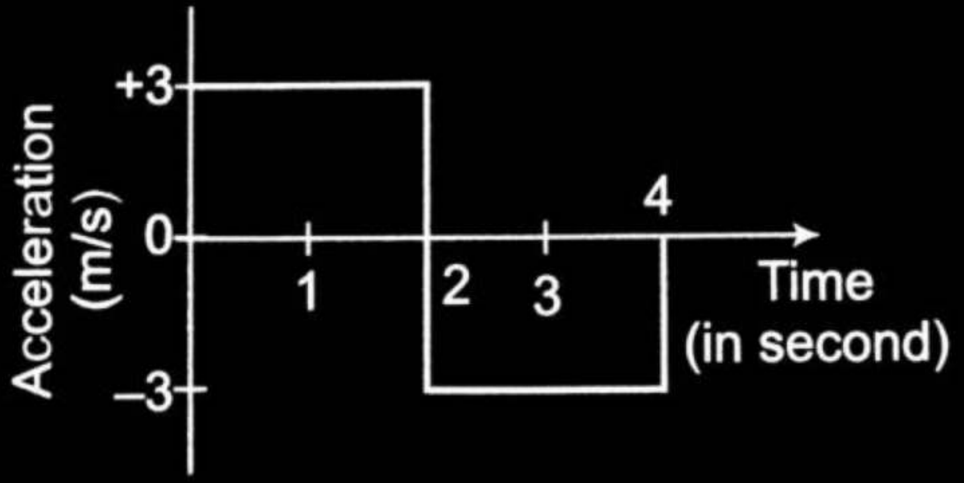


1. 110 m/s
2. 55 m/s
3. 550 m/s
4. 660 m/s

Question

67

A particle starts from rest at $t = 0$ and moves in a straight line with an acceleration as shown below. The velocity of the particle at $t = 3s$ is



1. 2 m/s
2. 3 m/s
3. 4 m/s
4. 6 m/s

Motion under gravity

Motion under gravity

↓ Special case of motn with cost^n accn.

→ Motion with Uniform acceleration

→ Non-Uniform motion, Variable Velocity

→ $a = g$ (air resistance ignore)

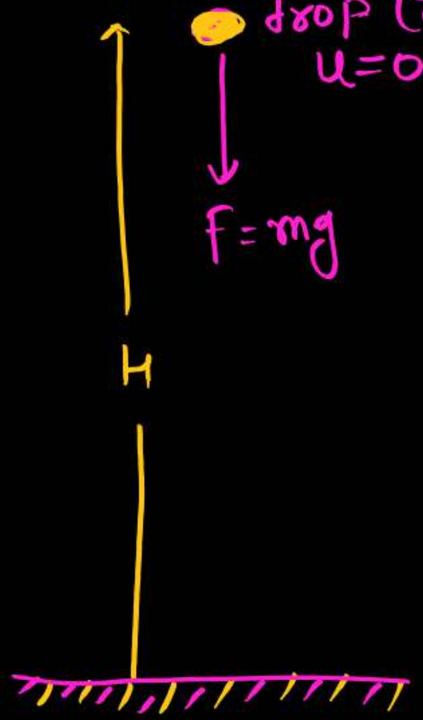
→ Jabtak question me air resistance given nahi ho use consider Nahi Karna hai

→ all eqn of motion is Valid here ✓

→ all case of consn accn is Valid here ✓

→ Jo question accn is not given then also consn $a = g = 10 m/s^2$

Case-1



$a = \frac{F}{m} = \frac{[mg]}{m}$

$a = g = 9.8 m/s^2 = cost^n$

$\approx 10 m/s^2$ ✓

= does not depend on mass

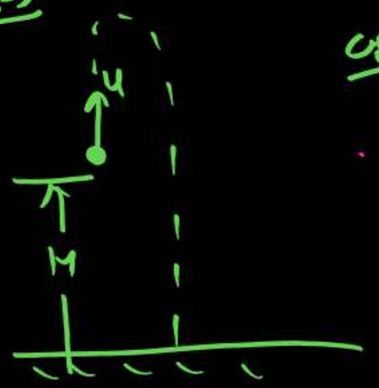
$S = H = \frac{1}{2} g t^2$

Same

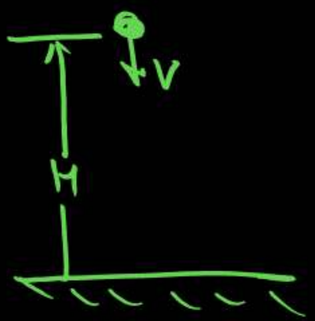
Case-2



Case-3



Case-4



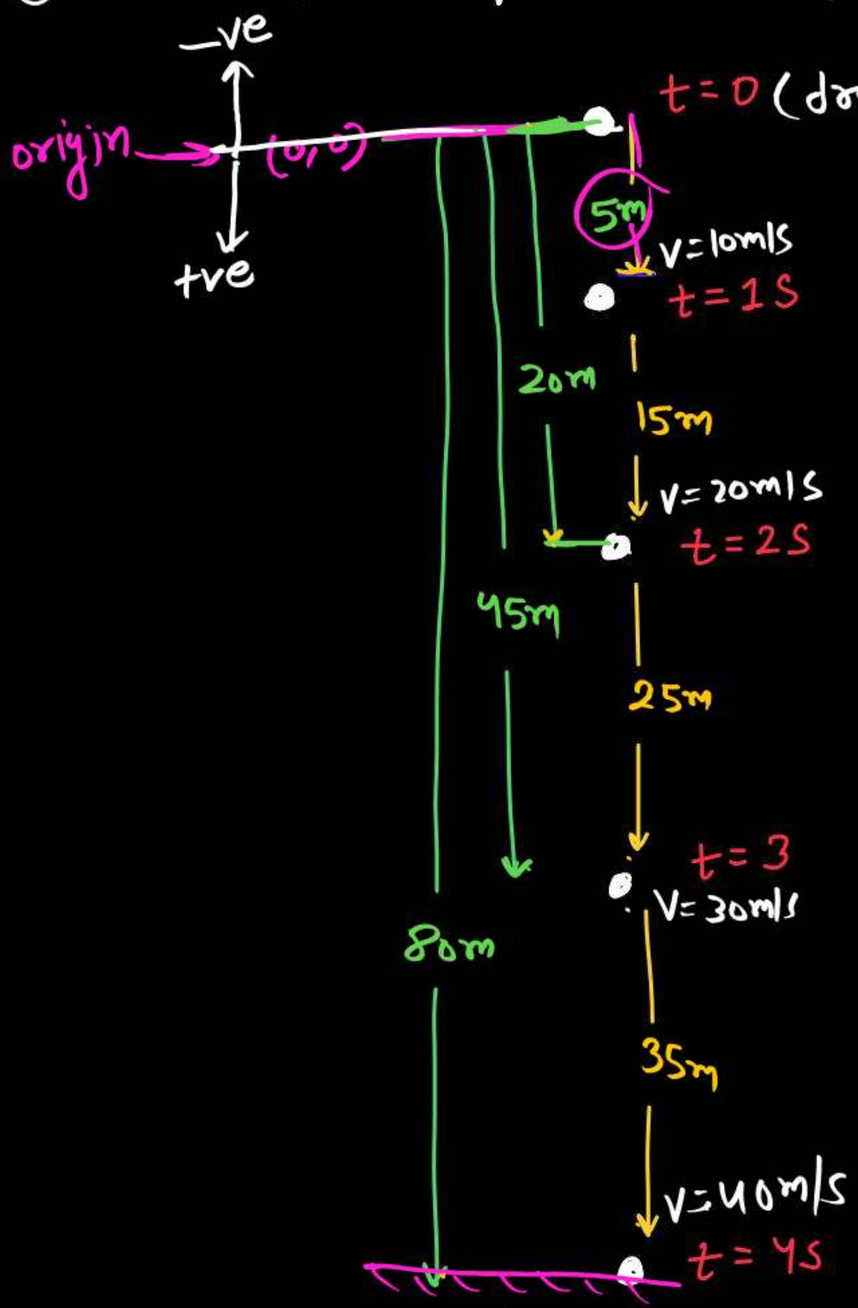
Question

No Copy

Motion under gravity is an example of

1. Non-uniform acceleration, uniform motion
2. Non-uniform motion, Non-uniform acceleration
3. Non-uniform motion, uniform acceleration
4. Uniform motion, uniform acceleration

① Object is droped ($u=0$)



$a = g$ [downward]

$a = +g = +10\ m/s^2$

for displacement in 't' sec

$$s = \cancel{ut}^{0} + \frac{1}{2}a(t)^2$$

$$s = \frac{1}{2}g(t)^2$$ ← Parabola.

$S_{1s} = \frac{1}{2}10(1)^2 = 5m$

$S_{2s} = \frac{1}{2}10(2)^2 = 20m$

$S_{3s} = \frac{1}{2}10(3)^2 = 45m$

$S_{4s} = \frac{1}{2}10(4)^2 = 80m$

$S_{5s} = \frac{1}{2}10(5)^2 = 125m$

$S_{6s} = 180m$

$S_{1s} : S_{2s} : S_{3s} : S_{4s} : S_{5s} = 5 : 20 : 45 : 80 : 125$

$= x : 4x : 9x : 16x : 25x$

$S_{1st} : S_{2nd} : S_{3rd} : S_{4th} = 5 : 15 : 25 : 35 : 45$

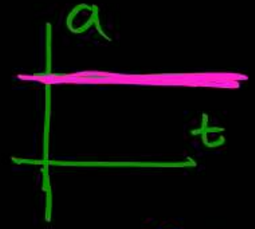
$= x : 3x : 5x : 7x : 9x$

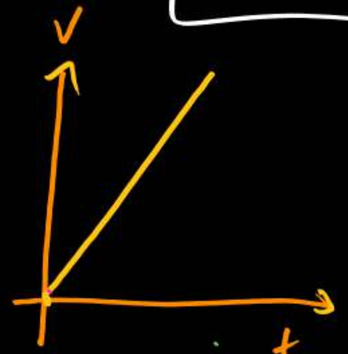
ratio of odd no

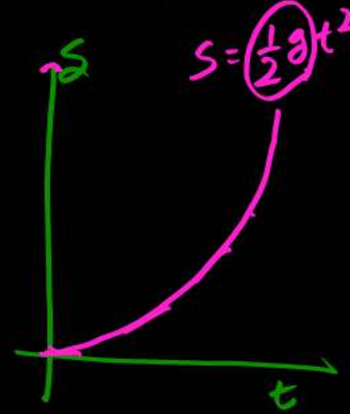
$v = \cancel{u}^{0} + at$ (velocity at t)

$v = 0 + gt$

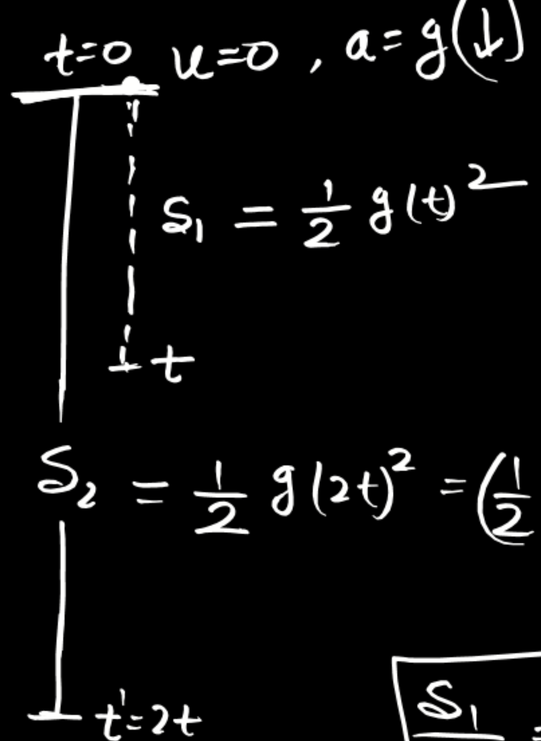
$v = 10t$







(2) object is dropped then find ratio of dispⁿ in time t & '2t'

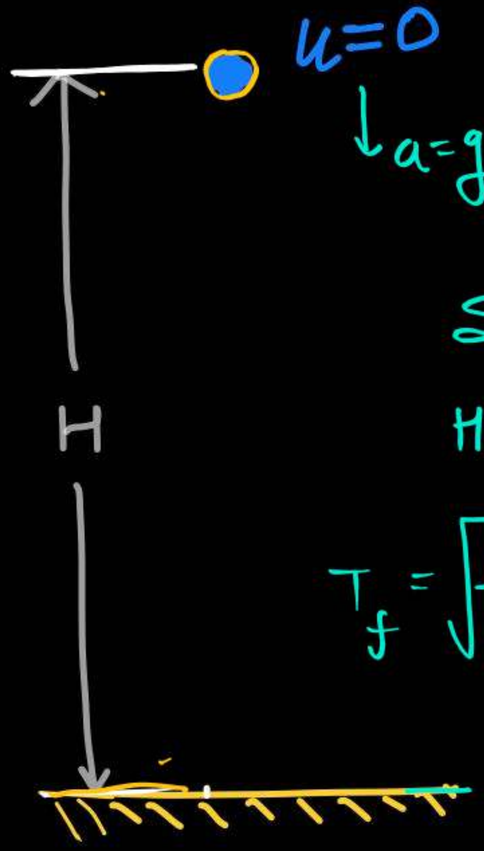


$$\frac{S_1}{S_2} = \frac{1}{4}$$

$$S_t : S_{next\ t} = x : 3x$$

(3)

Object is droped from height 'H' then find time taken to reach ground and velocity at ground.



$s = ut + \frac{1}{2}at^2$ ✓

$H = \frac{1}{2}gt^2$

$T_f = \sqrt{\frac{2H}{g}}$

$u_i = 0$
$a = g$
$s = H$
$v_f = ??$

3rd eqn of motion

$v^2 - u^2 = 2as$

$v^2 = 2gH$

$v = \sqrt{2gH}$ ✓

(5)

Ball is droped then find ratio of distance in 3rd sec and 7th sec

displament same here

Soln

$$S_{3rd\ sec} = u + \frac{a}{2}(2n-1)$$

$$S_{3rd} = 0 + \frac{10}{2}(2\times 3 - 1)$$

$$= 5(5) = 25m$$

$$S_{7h} = 0 + \frac{10}{2}(2\times 7 - 1)$$

$$= 65m$$

$$\frac{S_{5h}}{S_{7h}} = \frac{25}{65} = \frac{5}{13}$$

methad -2

ratio = 1 : 3 (5) 7 : 9 : 11 : (13) 5

$= \frac{5}{13}$ ✓

(5) Object droped from Height 180m from ground find its velocity at ground. and totel time of flight:-

done

$$T = \sqrt{\frac{2H}{g}}$$

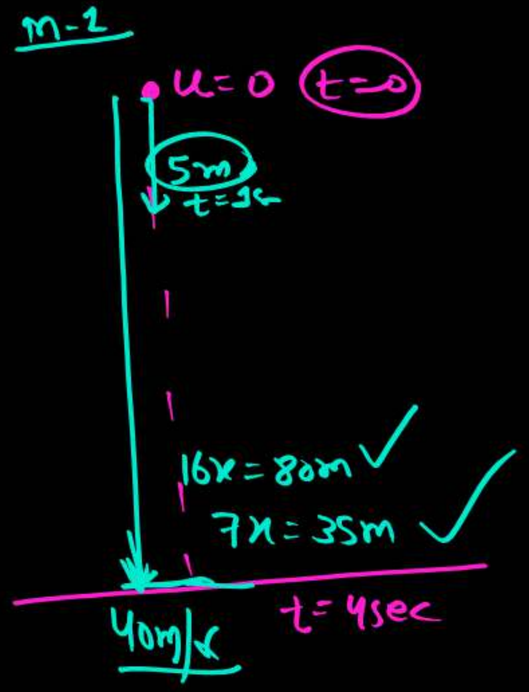
$$= \sqrt{\frac{2 \times 180}{10}}$$

$$= \sqrt{36}$$

$$= 6 \text{ sec}$$

$$u = 60 \text{ m/s}$$ ✓

Ball is droped and touches the ground in 4-sec. then find height of tower from it droped, velocity at ground, and distance travelled in last-sec of its Journey.



$u=0 \quad T_f = 4sec$

$S = ut + \frac{1}{2}at^2$

$= \frac{1}{2} \times 10 (4)^2$

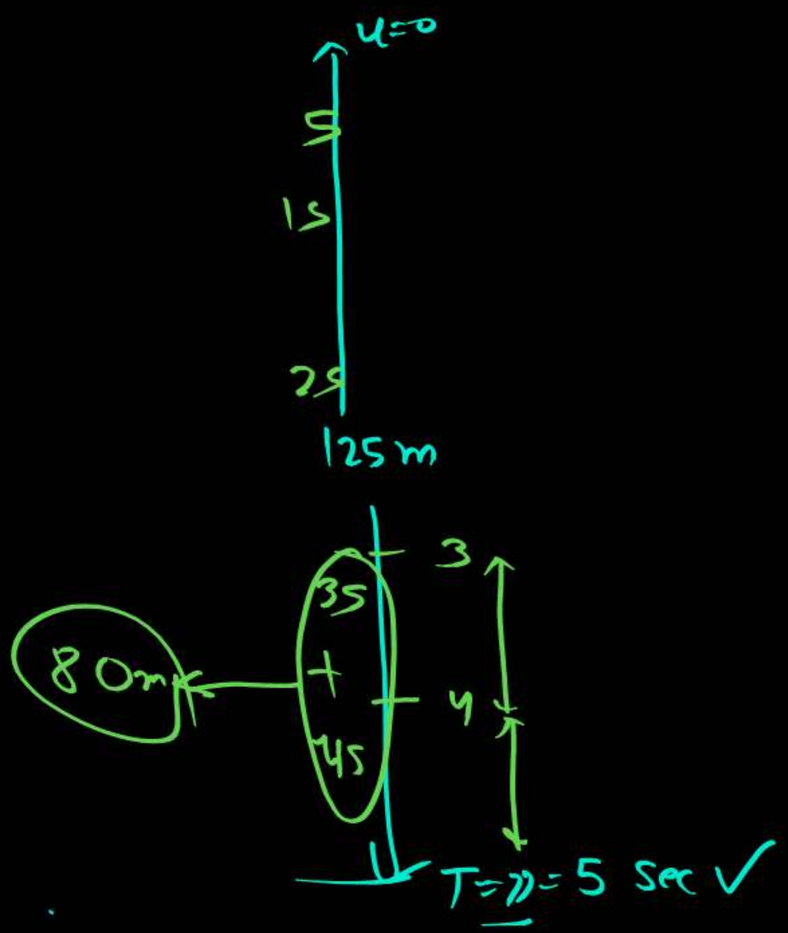
$= 5 \times 16 = 80m$

$V = \sqrt{2gh} = \sqrt{2 \times 10 \times 80} = \sqrt{16 \times 100}$

$= 4 \times 10$

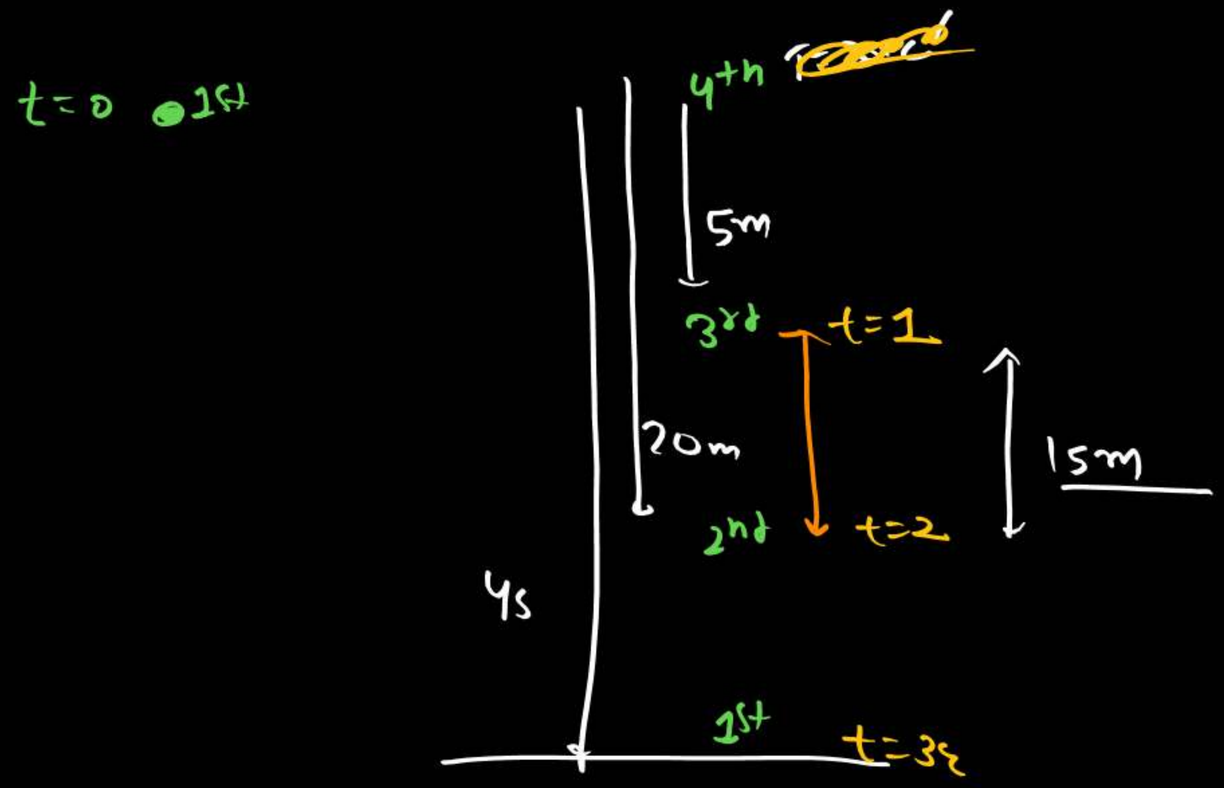
$V = u + at = 0 + 10 \times 4 = 40m/s$

Ball is droped from 125m then distance moved in last 2 sec of Journey.



Question

A ball is dropped at $t = 0$ sec after 1 sec 2nd ball is dropped after 2 sec 3rd ball is dropped, after 3 sec, 4th ball is dropped. Then, find distance between 2nd and 3rd ball when 4th ball is about to fall.



Ball is droped from some height and travel 55m in last-sec of its Journey then find time of flight and maxm height.

$S_{nth} = u + \frac{a}{2}(2n-1)$

$55 = 0 + \frac{10}{2}(2n-1)$

$11 = 2n-1$

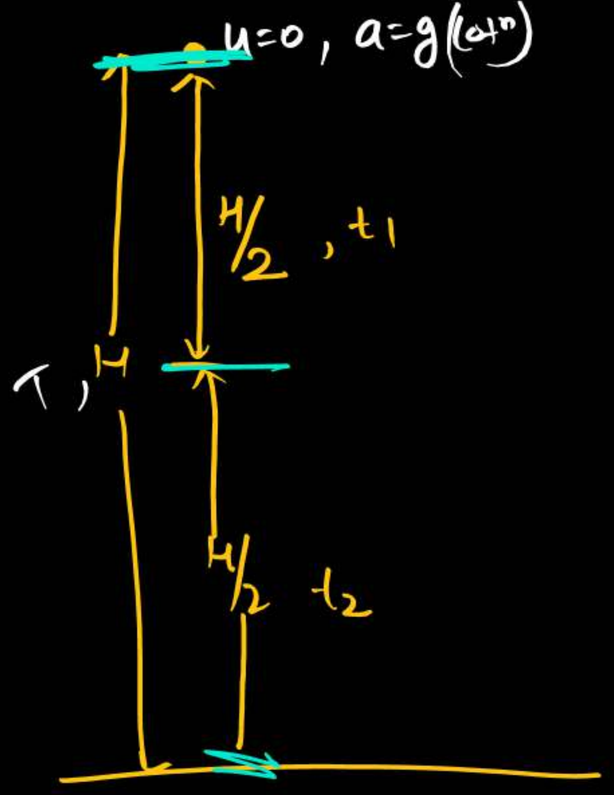
$12 = 2n$

$n = 6\,sec$

$S_{nth} = \frac{10}{2}(2n-1)$

$S_n = 10(n) - 5$

Ball is dropped from Height H then find ratio of time in 1st half and 2nd half distance.



$t_1 : t_2 = 1 : \sqrt{2}-1$ sec done ✓

$t_1 = \frac{T}{\sqrt{2}}$ —— done ✓

$t_2 = T - \frac{T}{\sqrt{2}}$

$= T\left(1 - \frac{1}{\sqrt{2}}\right)$

$= T\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$ ✓

- Object droped & it takes 1 sec to cover 2nd half height; then find time of flight.

$$t_2 = T\left(1-\frac{1}{\sqrt{2}}\right)$$

$$1 = T\left(1-\frac{1}{\sqrt{2}}\right)$$

$$1 = T\left(\frac{\sqrt{2}-1}{\sqrt{2}}\right)$$

$$T = \left[\frac{\sqrt{2}}{\sqrt{2}-1}\right] \text{sec}$$

$$T = \frac{\sqrt{2}}{\sqrt{2}-1} \times \frac{\sqrt{2}+1}{\sqrt{2}+1}$$

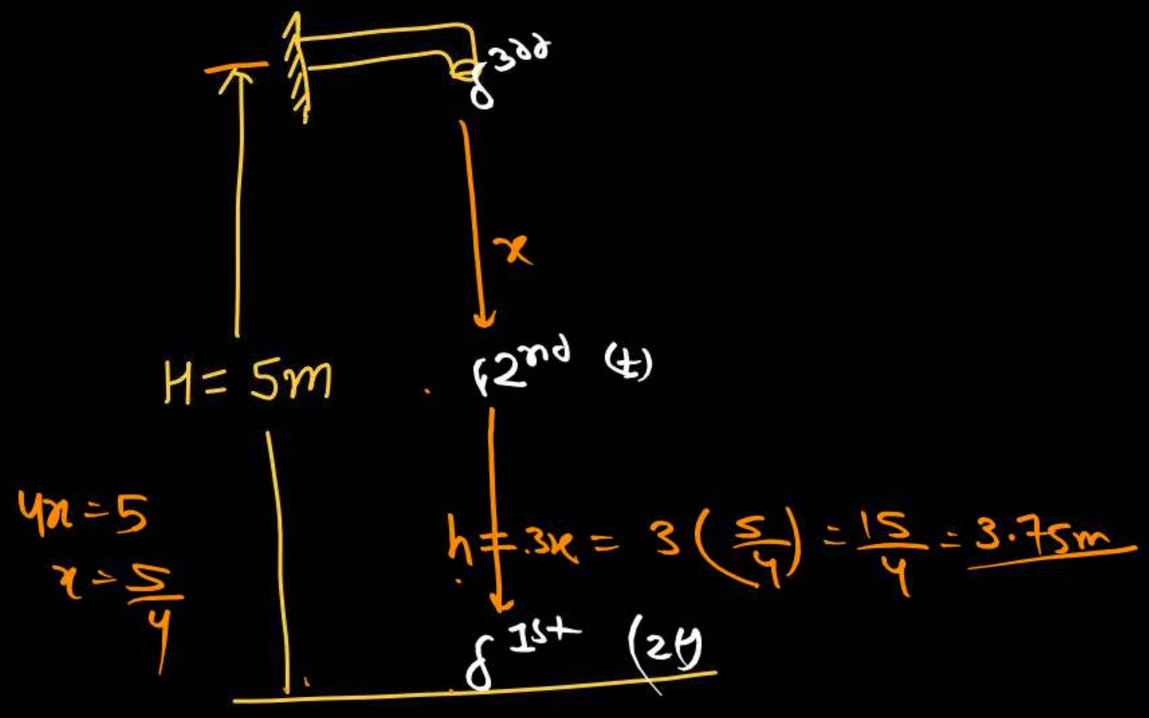
$$= \frac{\sqrt{2}(\sqrt{2}+1)}{(\sqrt{2})^2-(1)^2}$$

$$= \frac{2+\sqrt{2}}{1}$$

$$= (2+\sqrt{2})$$

Water drop is falling in regular intraval then find height of 2nd drop from ground if 3rd drop is about to fall when 1st just reach ground.

t = time gap



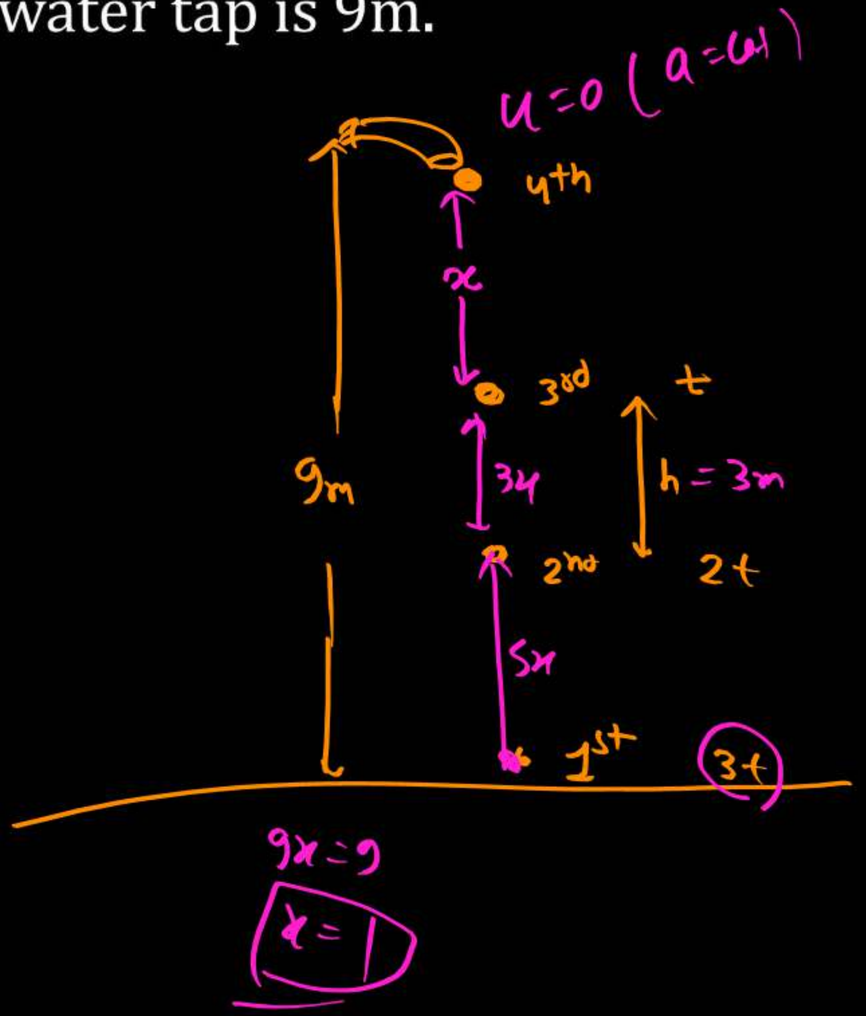
$4x = 5$

$x = \frac{5}{4}$

$h = 3x = 3\left(\frac{5}{4}\right) = \frac{15}{4} = 3.75m$

Question

Water drop is falling in a regular intervals when 1st drop is reaches to ground then 4th drop is about to release, then find distance between 2nd drop and 3rd drop. Height of water tap is 9m.



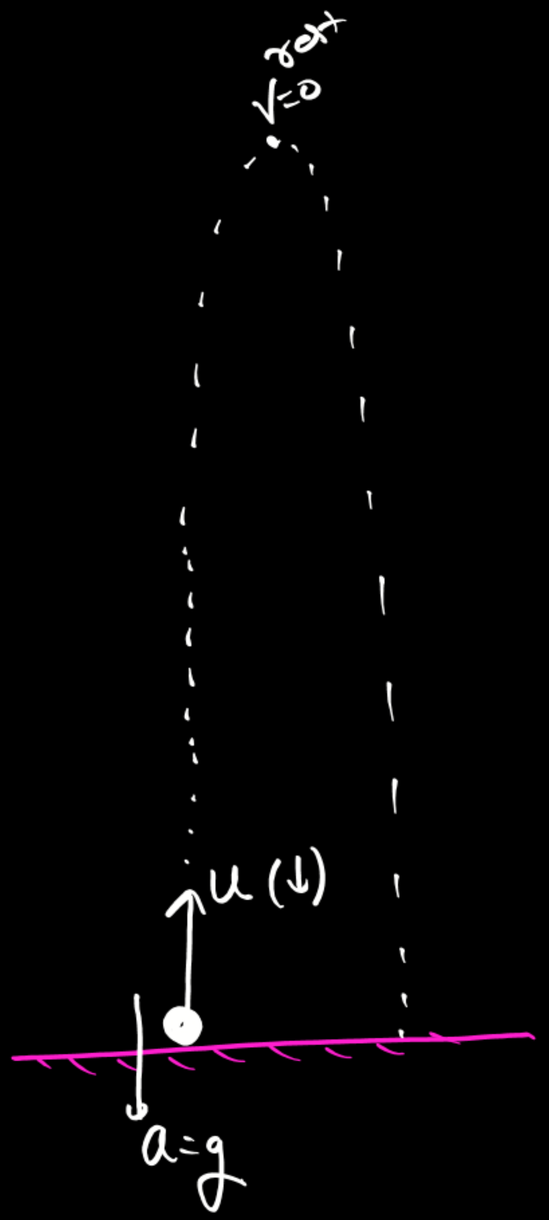
ground to ground motion under gravity

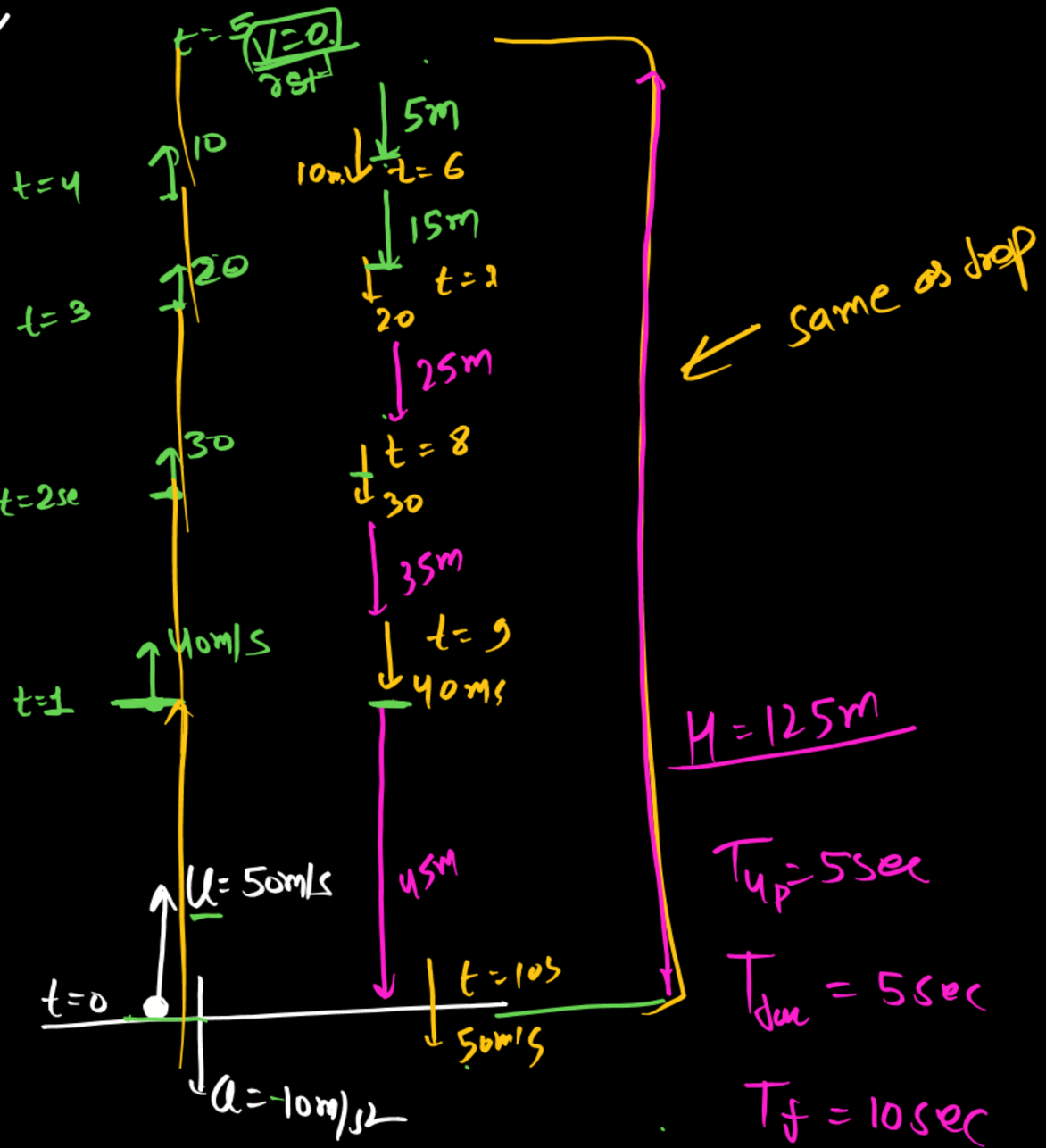
$V = u + at$

$= 50 - 10(2)$

$= 50 - 20$

$V = 30$





← same as drop

$H = 125m$

$T_{up} = 5sec$

$T_{dow} = 5sec$

$T_f = 10sec$

do all H/W in rough

Ball is droped from 80m then find ratio of distance moved in 1st 2sec and last 2-sec of Journey.

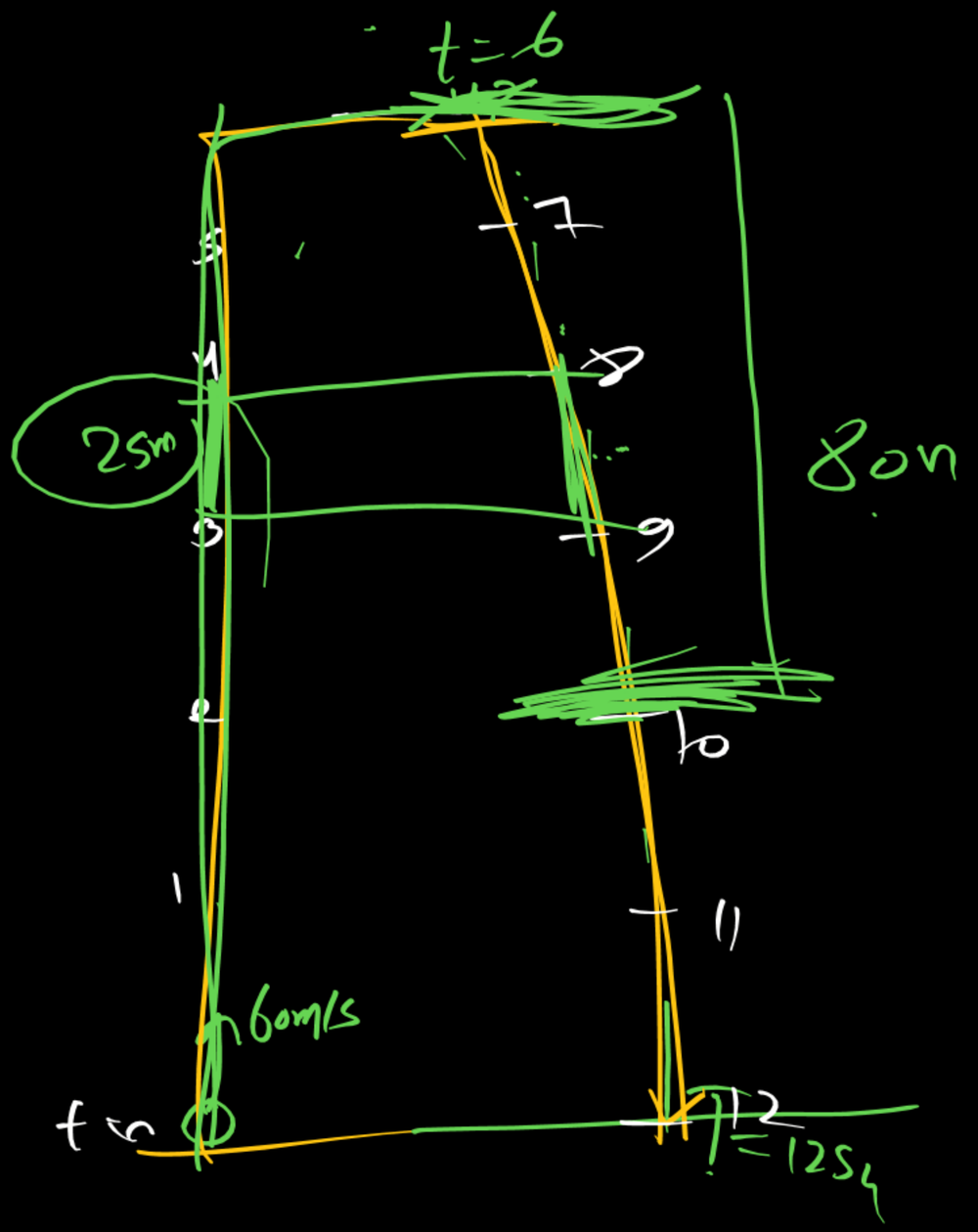
H/W

object is droped and distance in last 1sec is equal to 1st 3sec then find height from ground from where ball is droped

45m

H/W

$$S_{nth} = u + \frac{g}{2}(2n-1)$$



$T_{fly} = 12\ sec$

$H = 180\ m$ ✓

$S_{10\,sec\,(dist)} = 180 + 80\ m$

Ball is droped from height 125m after 3sec it stoped and relesed at same instant Find Total time of flight.

Ball is projected with speed 40 m/s then find

(i) H_{max}

(ii) T_f

(iv) Tupward

(v) speed after $t = 5$ sec, 6 sec

(vi) distance in 6 sec.

(vii) displacement in 6 sec

(viii) Avg speed in 7 sec

(ix) Avg velocity in 5 sec

(x) distance moved in 8th sec.

object is projected with 80m/s then find Avg. speed and velocity in 8-sec -

Object is projected with 40m/s and 60m/s respectively then find ratio of distance in last-sec of upward Journey.

A Ball is thrown upward with speed v_0 m/s then find Avg. velocity of upward Journey. and speed at half of the maximum Height.

A Ball is thrown upward with u_0 if its velocity at half of Maximum Height is 20 m/s then find it velocity u_0

Ball is droped from Height H and it travell $\frac{7H}{16}$ in last sec of Journey then find H'

Ball is projected up with 70m/s then find displacement in 10th sec and 10 sec.

Ball is projected up its position at $t = 7s$ and $t = 11s$ is same then find velocity of projection and max^m Height.

object is projected up with u its height at 3sec and 13 sec is same find u and that Height.

Ball is projected up its distance travelled in 3rd sec and 8th sec is same then find max height

Ball is projected up with 55 m/s then find distance in 6th sec.

Question

If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is **[2003]**

1. ut
2. $\frac{1}{2}gt^2$
3. $ut - \frac{1}{2}gt^2$
4. $(u + gt)t$

Question

The ratio of the distance traveled by a freely falling body in the 1st, 2nd, 3rd and 4th second:

[MR* (2022)]

1. 1 : 1 : 1 : 1
2. 1 : 2 : 3 : 4
3. 1 : 4 : 9 : 16
4. 1 : 3 : 9 : 16

Question

A ball is thrown vertically downward with a velocity of 20 m/s from the top of a tower. It hits the ground after some time with a velocity of 80 m/s. The height of the tower is: (g = 10 m/s^2) **[2020]**

1. 340 m
2. 320 m
3. 300 m
4. 360 m

Question

A stone falls freely under gravity. It covers distances h_1, h_2 and h_3 in the first 5 seconds, the next 5 seconds and the next 5 seconds respectively. The relation between h_1, h_2 and h_3 is: **(2013)**

1. $h_1 = h_2 = h_3$
2. $h_1 = 2\,h_2 = 3\,h_3$
3. $h_1 = \frac{h_2}{3} = \frac{h_3}{5}$
4. $h_2 = 3\,h_1$ and $h_3 = 3\,h_2$

Question

A boy standing at the top of a tower of 20 m height drops a stone. Assuming $g = 10\ m/s^2$, the velocity with which it hits the ground is: **(2011 Pre)**

1. 10.0 m/s
2. 20.0 m/s
3. 40.0 m/s
4. 5.0 m/s

Question

A ball is dropped from a high rise platform at $t = 0$ starting from rest. After 6 seconds another ball is thrown downwards from the same platform with a speed v. The two balls meet at $t = 18$ s. What is the value of v? **(2010 Pre)**

1. 75 m/s
2. 55 m/s
3. 40 m/s
4. 60 m/s

Question

Two bodies, A (of mass 1 kg) and B (of mass 3 kg) are dropped from heights of 16 m and 25 m, respectively. The ratio of the time taken by them to reach the ground is: **(2006)**

1. 5/4
2. 8/5
3. 5/8
4. 4/5

Question

If a ball is thrown vertically upwards with speed u, the distance covered during the last t seconds of its ascent is [2003]

1. ut
2. $\frac{1}{2}gt^2$
3. $ut - \frac{1}{2}gt^2$
4. $(u + gt)t$

Question

A man throws ball with the same speed vertically upwards one after the other at an interval of 2 seconds. What should be the speed of the throw so that more than two balls are in the sky at any time? (Given $g = 9.8\ m/s^2$) **[MR* (2003)]**

1. More than 19.6 m/s
2. At least 9.8 m/s
3. Any speed less than 19.6 m/s
4. Only with speed 19.6 m/s

Question

A particle is thrown vertically upward. Its velocity at half of the height is 10 m/s, then the maximum height attained by it: (g = 10 m/s^2) **(2001)**

1. 8 m
2. 20 m
3. 10 m
4. 16 m

Question

A body starts falling from height 'h' and travels distance $h/2$ during last second of motion then time of flight is (in second): **(1999)**

1. $\sqrt{2}-1$
2. $2+\sqrt{2}$
3. $\sqrt{2}+\sqrt{2}$
4. $\sqrt{3}+2$

Question

A body dropped from a height h with initial velocity zero, strikes the ground with a velocity 3 m/s. Another body of same mass dropped from the same height h with an initial velocity of 4 m/s. The final velocity of second mass, with which it strikes the ground is:

[MR* (1996)]

1. 5 m/s
2. 12 m/s
3. 3 m/s
4. 4 m/s

Question

The water drop falls at regular intervals from a tap 5 m above the ground. The third drop is leaving the tap at instant the first drop touches the ground. How far above the ground is the second drop at that instant? **(1995)**

1. 3.75 m
2. 4.00 m
3. 1.25 m
4. 2.50 m

Thank You